



# ODATANO

ODATA FOR CARDANO

THE BRIDGE BETWEEN ENTERPRISE SYSTEMS LIKE SAP AND CARDANO

## THE STARTING POINT

# Translating between ERP Systems and Cardano

## Enterprise Systems / SAP

Where the back office of the global economy actually runs. From purchasing, to finance, supply chain, etc.

# 400K+

SAP CUSTOMERS WORLDWIDE



## Cardano

UTXO blockchain with native assets and Plutus smart contracts. Backend access via Nodes, Blockfrost, Koios, Ogmios etc.

# 3

NETWORKS · PREVIEW · PREPROD · MAINNET

## INTEGRATION EFFORT TODAY

# Connecting SAP to Cardano requires custom work in five areas.

- 01 Learn Cardano-specific APIs and CBOR serialization.
- 02 Manage Blockfrost, Koios, Ogmios providers and failover logic.
- 03 Build custom caches, indexers and rate limiters.
- 04 Hand-roll coin selection, fee estimation and signing flows.
- 05 Glue all of the above into Fiori, ABAP and BTP workflows.

**The Solution:** Hiding the complexity of the Cardano Blockchain behind a protocol that Enterprise Systems are already familiar with → OData

OData (Open Data Protocol) is an ISO/IEC approved, OASIS standard  
OASIS (Organization for the Advancement of Structured Information Standards) SAP / IBM / Microsoft etc.

ODATANO exposes the Cardano blockchain as a standard OData V4 service.

## WHAT IS ODATA?

# OData - A standard query protocol for data over HTTP

OData turns REST into a true standard for enterprise Systems

**Open standard:** OASIS- and ISO/IEC-certified protocols

**Self-describing:** Machine-readable metadata exposes the full data model

**Plug-and-play:** Generic clients and tools work out of the box. No protocol expertise needed

EXAMPLE — READ A WALLET BALANCE

```
GET /odata/v4/cardano/Addresses(  
  'addr1q9...')  
?$select=balance, stake  
&$expand=utxos  
  
// returns OData JSON  
{  
  "balance": 924380000,  
  "stake": "stake1u...",  
  "utxos": [ ... ]  
}
```

## WHY THIS MATTERS FOR CARDANO

A large blue '1m+' logo, where the '1' and 'm' are in a bold, sans-serif font, and the '+' is a smaller blue plus sign.

businesses that use OData worldwide  
everyday

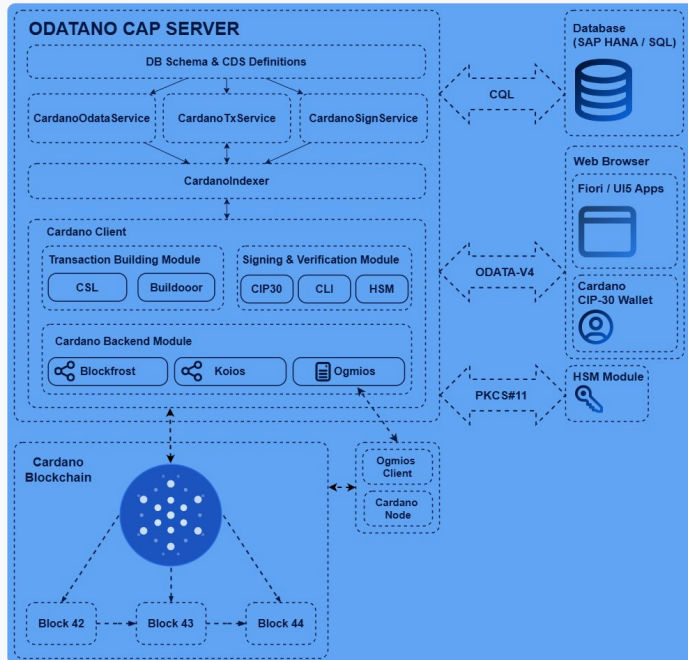
## OData V4 is a known interface in enterprise IT.

ERP developers already use OData. **ODATANO exposes Cardano through the same interface, without requiring blockchain-specific code on the consumer side.**

Not just SAP: OData V4 is an OASIS / ISO standard. The API is consumable by any OData client. SAP, Microsoft, Salesforce, custom clients etc.

## ARCHITECTURE

# CAP server with three services and a multi-backend client.



1

## Three CAP services

Odata Service for reads, TxService for transactions, SignService for external signers, all defined in CDS.

2

## Multi-backend client

Blockfrost, Koios and Ogmios behind one interface, with automatic failover and TTL-based caching.

3

## Two transaction builders

CSL or Buildooor. Coin selection, fees and Plutus V3 are handled internally.

4

## Three signing paths

CIP-30 browser wallets, CLI keys, or HSM via PKCS#11 for production-grade key custody.



# Read · Build · Sign · Submit.

## STEP 01

### Read

29 CDS entities and 15 read actions over standard OData filters.

```
GET/Transactions('0x...')
```

## STEP 02

### Build

8 transaction-build actions  
ADA, native assets, Plutus  
Contract interactions etc.

```
POST /BuildAdaTransfer
```

## STEP 03

### Sign

CIP-30, CLI or HSM. State  
machine: pending →  
verified.

```
POST /VerifySignature
```

## STEP 04

### Submit

Broadcast to the network  
and return the transaction  
hash.

```
POST /SubmitTransaction
```

# TRACE

## Supply chain custody on-chain

Reference implementation built on ODATANO.

Custody handovers are NFT transfers with all transfer rules are Plutus validators.

1. Mint a custody NFT per shipment with inline datums.
2. Transfer custody when validator conditions are met.
3. Auditors query the chain via standard OData filters.
4. No custom blockchain code in the host SAP system.
5. All Blockchain Calls routed over ODATANO

TRACE

Batches

Participants

✓ Verify

Verify Batch

asset13g7ep7hrtn0agemnyqlwvgeyviewzlgfqqwefu4

Verify

Fingerprint: asset13g7ep7hrtn0agemnyqlwvgeyviewzlgfqqwefu4  
Current Holder (vkh): d090525914fb9bcd35141eaff7b054b9ce105f154ebb73347f9c741  
Custody Step: 2

Chain of Custody

|                            |                                                       |          |   |
|----------------------------|-------------------------------------------------------|----------|---|
| ✓ Step 0 — MINT            | Holder: d090525914fb9bcd...   Tx: 7f638c8952946aff... | verified | > |
| ✓ Step 1 — TRANSFER        | Holder: d090525914fb9bcd...   Tx: 42537e111656f9f3... | verified | > |
| ✓ Step 2 — TRANSFER        | Holder: d82fd3efe829abf5...   Tx: 778e4cd89e43678c... | verified | > |
| ✓ Step 3 — DOCUMENT_ANCHOR | Holder: d82fd3efe829abf5...   Tx: e9862cba7d1992ea... | verified | > |
| ✓ Step 4 — DOCUMENT_ANCHOR | Holder: d82fd3efe829abf5...   Tx: 7358c4dfd876d63a... | verified | > |
| ✓ Step 5 — RECALL          | Holder: d82fd3efe829abf5...   Tx: c4e8746cb1337f53... | verified | > |

**QUANTIX**

## On-chain order commitments with oracle-attested data

Buyer picks a supplier and a quantity. One atomic Cardano transaction mints a receipt NFT, conditional on Charli3 oracle inventory and price feeds.

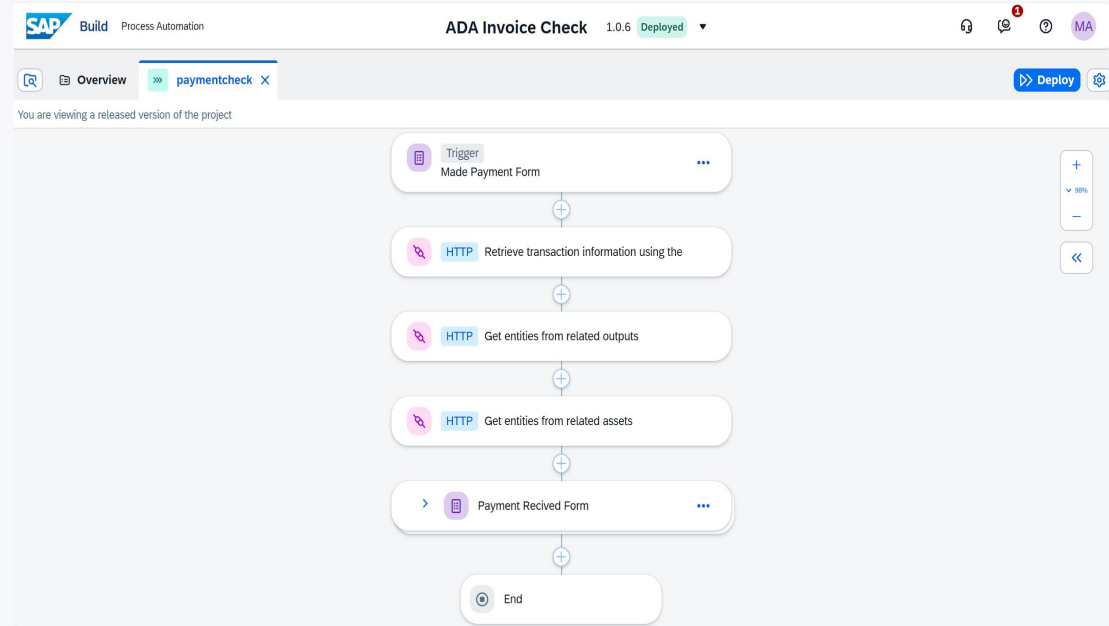
1. SAP CAP buyer app; ODATANO as the chain plugin.
2. Forked Charli3 stack with multi-feed inventory + price oracles.
3. Custom Plutus V3 minting policy enforces the trade at execution time.
4. First place at the Charli3 Catalyst Hackathon, April 2026.

| QUANTIX   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| My Buys   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Suppliers |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Feeds     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|           | <div>Oracle Feeds</div> <div> <div> <div>Lithium GmbH — Warehouse Feed</div> <div> <div>11.336 g</div> <div>Freshness</div> <div>8m left</div> <div>Last on-chain update</div> <div>2m ago (12:44:36)</div> <div>Coordinator wallet (data authority)</div> <div><a href="#">addr_test1vqk5...3w89t2</a></div> <div>Oracle asset</div> <div>C3AS_inventory</div> <div>Oracle policy</div> <div>78adae0debdf47...7ec622</div> </div> </div> <div> <div>Lithium GmbH — Pricing Desk Feed</div> <div> <div>10.00 ADA/g</div> <div>Freshness</div> <div>8m left</div> <div>Last on-chain update</div> <div>2m ago (12:44:36)</div> <div>Coordinator wallet (data authority)</div> <div><a href="#">addr_test1vrxk...nv092g</a></div> <div>Oracle asset</div> <div>C3AS_price</div> <div>Oracle policy</div> <div>78adae0debdf47...7ec622</div> </div> </div> <div> <div>Cobalt AG — Warehouse Feed</div> <div> <div>67.585 g</div> <div>Freshness</div> <div>expired</div> <div>Last on-chain update</div> <div>19. Apr., 10:58</div> <div>Coordinator wallet (data authority)</div> <div><a href="#">addr_test1vzz4...2y6zh</a></div> <div>Oracle asset</div> <div>C3AS_inventory</div> <div>Oracle policy</div> <div>7e3cc787cd71c6...bd081f</div> </div> </div> </div>     |
|           | <div>Oracle Feeds</div> <div> <div> <div>Cobalt AG — Pricing Desk Feed</div> <div> <div>7.86 ADA/g</div> <div>Freshness</div> <div>expired</div> <div>Last on-chain update</div> <div>19. Apr., 10:58</div> <div>Coordinator wallet (data authority)</div> <div><a href="#">addr_test1vp4s...f9cpjq</a></div> <div>Oracle asset</div> <div>C3AS_price</div> <div>Oracle policy</div> <div>7e3cc787cd71c6...bd081f</div> </div> </div> <div> <div>Rare-Earth Ltd. — Warehouse Feed</div> <div> <div>75.471 g</div> <div>Freshness</div> <div>expired</div> <div>Last on-chain update</div> <div>19. Apr., 10:45</div> <div>Coordinator wallet (data authority)</div> <div><a href="#">addr_test1vprk...7rytzi</a></div> <div>Oracle asset</div> <div>C3AS_inventory</div> <div>Oracle policy</div> <div>dc8ffa5e4ed0ae...a37ed9</div> </div> </div> <div> <div>Rare-Earth Ltd. — Pricing Desk Feed</div> <div> <div>27.35 ADA/g</div> <div>Freshness</div> <div>expired</div> <div>Last on-chain update</div> <div>19. Apr., 10:45</div> <div>Coordinator wallet (data authority)</div> <div><a href="#">addr_test1vzhn...nigsoc</a></div> <div>Oracle asset</div> <div>C3AS_price</div> <div>Oracle policy</div> <div>dc8ffa5e4ed0ae...a37ed9</div> </div> </div> </div> |

# Cardano as a data source in SAP Build.

ODATANO registers as a standard OData V4 destination on SAP BTP. From there, SAP Build Apps, Process Automation and Work Zone consume it like any other enterprise service.

1. Bind Cardano entities to Build pages without writing code.
2. Trigger build/sign/submit actions from Process Automation flows.
3. Surface wallets, transactions and NFTs as Work Zone tiles.
4. Authentication and roles via XSUAA. Same as any BTP app.



## BY THE NUMBERS

# Test coverage and performance.

**1,285**

AUTOMATED TESTS

**97** %

STATEMENT COVERAGE

**<5** ms

AVG WARM RESPONSE

**96.5** %CACHE SPEEDUP COLD →  
WARM**29**

CDS ENTITIES

**23**

ODATA ENDPOINTS

**3**BACKENDS · BLOCKFROST ·  
KOIOS · OGMIOS**10** +

DOCUMENTATION GUIDES

# Four milestones · 5 Months · Fully delivered and approved.

**M1 · JAN 2026**

## **Read operations**

18 entities, 15 read actions, multi-backend with failover.

✓ **APPROVED**

**M2 · FEB 2026**

## **Transaction building**

Build → Sign → Submit.  
CSL + Buildoor. Plutus V3.

✓ **APPROVED**

**M3 · MAR 2026**

## **External signing**

CIP-30, CLI, HSM (PKCS#11).  
Wallet Viewer + ABAP.

✓ **APPROVED**

**M4 · APR 2026**

## **Close-out**

Apache 2.0, npm, full docs, demo videos, TRACE demo.

✓ **APPROVED**

|       |                                           |                                                                                        |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------|
| NEXT  | <b>More Use Case Apps</b>                 | More reference apps like QUANTIX or TRACE / library for different USE Cases of ODATANO |
| NEAR  | <b>Public Hosted ODATANO</b>              | Managed multi-tenant deployment for easy “try out” solutions for enterprises           |
| NEAR  | <b>Deeper BTP alignment</b>               | Destinations, Workflow + Work Zone tiles, packaged SAP Build templates.                |
| LATER | <b>First real enterprise integrations</b> | First integrations in existing ERP Landscapes / Listing on the SAP App Store           |

# Thank you.

Apache 2.0 · funded by Cardano Catalyst Fund 14 · built on SAP CAP.

**WEB**`odatano.dev`**GITHUB**`github.com/ODATANO/ODATANO`**NPM**`@odatano/core`**DOCS**`/docs/QUICK_START.md`**CATALYST**`Fund 14 · Project 1400109`**CONTACT**`max@maxalexweber.de`